Advanced Engineering Mathematics

Fourier Series

Engr. Kaiveni Tom Dagcuta
Computer Engineering

1 Fourier Series

Key Concept

Fourier series represent periodic functions as sums of simple sinusoidal functions. The central idea is that a sufficiently well-behaved periodic function can be decomposed into harmonically related sine and cosine waves.

For a function with period $2L$, the Fourier series is written as

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right].$$

Fourier series extend the idea of representing a complicated object using simpler building blocks. In vector spaces, we express a vector as a combination of basis vectors. In Fourier analysis, we express a periodic function as a combination of orthogonal basis functions:

$$1, \quad \cos\left(\frac{\pi x}{L}\right), \quad \sin\left(\frac{\pi x}{L}\right), \quad \cos\left(\frac{2\pi x}{L}\right), \quad \sin\left(\frac{2\pi x}{L}\right), \quad \ldots$$

The connection to previous topics is important. From vectors and inner products, we know that orthogonality allows clean decomposition. From series, we know that infinite sums can approximate functions. Fourier series combines both ideas.

Engineering Note

In engineering, Fourier series are used to analyze vibrations, sound waves, alternating current signals, heat conduction, communication signals, filters, and periodic mechanical motion. Instead of studying a complicated waveform directly, we study its frequency components.

1.1 Trigonometric Fourier Series

Motivation and Intuition

Many engineering signals repeat. Examples include AC voltage, rotating machinery vibration, periodic forcing functions, sound waves, and repeated pulse trains.

A periodic function may look complicated in the time domain, but it can often be understood as a combination of a constant component, a fundamental sinusoid, and higher harmonics.

Key Concept

If $f(x)$ has period $2L$, its trigonometric Fourier series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) \right],$$

where

$$a_0 = \frac{1}{L}\int_{-L}^{L} f(x)\,dx,$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)\,dx, \qquad n = 1, 2, 3, \ldots$$

and

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)\,dx, \qquad n = 1, 2, 3, \ldots$$

Geometric Interpretation

The coefficients a_n and b_n measure how much of each cosine and sine wave is present in $f(x)$. This is similar to projecting a vector onto basis vectors.

The constant term $\frac{a_0}{2}$ represents the average or DC component of the function.

Key Concept

The functions

$$1, \quad \cos\left(\frac{n\pi x}{L}\right), \quad \sin\left(\frac{n\pi x}{L}\right)$$

are orthogonal on $[-L, L]$. This orthogonality is what allows each Fourier coefficient to be computed independently.

Key Properties

If $f(x)$ satisfies Dirichlet-type conditions on $[-L, L]$, then its Fourier series converges to

$$f(x)$$

at points where f is continuous, and to

$$\frac{f(x^+) + f(x^-)}{2}$$

at jump discontinuities.

Engineering Note

At a discontinuity, the Fourier series does not choose the left-hand value or the right-hand value. It converges to the average of the two limiting values. This is important when analyzing square waves, pulse signals, and switching circuits.

Worked Example

Example

Find the Fourier series of

$$f(x) = x, \qquad -\pi < x < \pi.$$

Here, the period is 2π, so $L = \pi$. The Fourier series has the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx)\right].$$

First compute a_0:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} x \, dx.$$

Since x is an odd function,

$$\int_{-\pi}^{\pi} x \, dx = 0.$$

Thus,

$$a_0 = 0.$$

Next compute a_n:

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos(nx) \, dx.$$

Since x is odd and $\cos(nx)$ is even, their product is odd:

$$x \cos(nx) \text{ is odd.}$$

Therefore,

$$a_n = 0.$$

Now compute b_n:

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin(nx) \, dx.$$

Since x is odd and $\sin(nx)$ is odd, their product is even:

$$x\sin(nx) \text{ is even.}$$

Thus,

$$b_n = \frac{2}{\pi}\int_0^\pi x\sin(nx)\,dx.$$

Use integration by parts:

$$u = x, \qquad dv = \sin(nx)\,dx.$$

Then

$$du = dx, \qquad v = -\frac{\cos(nx)}{n}.$$

So

$$\int x\sin(nx)\,dx = -\frac{x\cos(nx)}{n} + \frac{1}{n}\int \cos(nx)\,dx.$$

Therefore,

$$\int_0^\pi x\sin(nx)\,dx = \left[-\frac{x\cos(nx)}{n} + \frac{\sin(nx)}{n^2}\right]_0^\pi.$$

Evaluate at the endpoints:

$$= -\frac{\pi\cos(n\pi)}{n} + \frac{\sin(n\pi)}{n^2} - \left(0 + \frac{\sin(0)}{n^2}\right).$$

Since

$$\cos(n\pi) = (-1)^n, \qquad \sin(n\pi) = 0,$$

we get

$$\int_0^\pi x\sin(nx)\,dx = -\frac{\pi(-1)^n}{n}.$$

Hence,

$$b_n = \frac{2}{\pi}\left[-\frac{\pi(-1)^n}{n}\right] = \frac{2(-1)^{n+1}}{n}.$$

Thus,

$$x = \sum_{n=1}^{\infty} \frac{2(-1)^{n+1}}{n}\sin(nx), \qquad -\pi < x < \pi.$$

Engineering Insight

Engineering Note

The function $f(x) = x$ over $(-\pi, \pi)$ periodically extends into a sawtooth wave. Sawtooth waves occur in signal generation, oscillators, scanning circuits, and waveform synthesis. Its Fourier series contains only sine terms because the waveform is odd.

Common Mistakes and Notes

Key Concept

Common mistakes:

1. Forgetting that a_0 is divided by 2 in the Fourier series.
2. Using the wrong interval length. If the period is $2L$, use $\frac{n\pi x}{L}$.
3. Ignoring symmetry. Even and odd properties can greatly reduce computation.
4. Expecting the series to match the function exactly at jump discontinuities.

2 Fourier Cosine Series

Motivation and Intuition

If a function is even, then it is symmetric about the vertical axis. Since cosine functions are also even, an even function can be represented using only cosine terms.

This reduces the work needed to compute the Fourier series.

Key Concept

If $f(x)$ is even on $[-L, L]$, then

$$b_n = 0$$

for all n, and the Fourier series becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right),$$

where

$$a_0 = \frac{2}{L}\int_0^L f(x)\, dx,$$

and

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\, dx.$$

Geometric and Physical Interpretation

A Fourier cosine series represents an even periodic extension of a function. If the original function is defined only on $0 < x < L$, then choosing a cosine series reflects it evenly across the vertical axis.

$$f(-x) = f(x)$$

This is useful when the physical problem has symmetry, such as temperature distribution in a rod with symmetric boundary behavior.

Worked Example

Example

Find the Fourier cosine series of

$$f(x) = x, \qquad 0 < x < \pi.$$

For a half-range cosine expansion on $(0, \pi)$, we take $L = \pi$ and write

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(nx).$$

Compute a_0:

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x \, dx.$$

Thus,

$$a_0 = \frac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \frac{2}{\pi} \cdot \frac{\pi^2}{2} = \pi.$$

So,

$$\frac{a_0}{2} = \frac{\pi}{2}.$$

Now compute a_n:

$$a_n = \frac{2}{\pi} \int_0^{\pi} x \cos(nx) \, dx.$$

Use integration by parts:

$$u = x, \qquad dv = \cos(nx) \, dx.$$

Then

$$du = dx, \qquad v = \frac{\sin(nx)}{n}.$$

Therefore,

$$\int x \cos(nx) \, dx = \frac{x \sin(nx)}{n} - \frac{1}{n} \int \sin(nx) \, dx.$$

Since

$$\int \sin(nx) \, dx = -\frac{\cos(nx)}{n},$$

we get

$$\int x \cos(nx) \, dx = \frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2}.$$

Evaluate:

$$\int_0^{\pi} x \cos(nx) \, dx = \left[\frac{x \sin(nx)}{n} + \frac{\cos(nx)}{n^2} \right]_0^{\pi}.$$

At $x = \pi$:

$$\frac{\pi \sin(n\pi)}{n} + \frac{\cos(n\pi)}{n^2} = \frac{(-1)^n}{n^2}.$$

At $x = 0$:

$$0 + \frac{\cos(0)}{n^2} = \frac{1}{n^2}.$$

Hence,

$$\int_0^{\pi} x \cos(nx) \, dx = \frac{(-1)^n - 1}{n^2}.$$

Thus,

$$a_n = \frac{2}{\pi} \cdot \frac{(-1)^n - 1}{n^2}.$$

Therefore,

$$x = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2\left[(-1)^n - 1\right]}{\pi n^2} \cos(nx), \qquad 0 < x < \pi.$$

Since $(-1)^n - 1 = 0$ for even n and -2 for odd n,

$$x = \frac{\pi}{2} - \frac{4}{\pi} \sum_{m=1}^{\infty} \frac{\cos\left((2m-1)x\right)}{(2m-1)^2}.$$

Engineering Insight

Engineering Note

Fourier cosine series are especially useful in heat conduction problems on a finite rod when the boundary conditions naturally produce even extensions. They are also closely related to cosine transforms used in image compression and signal processing.

Common Mistakes and Notes

Key Concept

A Fourier cosine series does not mean the original function must already be even on $(0, L)$. It means we choose to extend the function evenly to $(-L, L)$.

3 Fourier Sine Series

Motivation and Intuition

If a function is odd, then it is symmetric through the origin. Since sine functions are odd, an odd function can be represented using only sine terms.

Key Concept

If $f(x)$ is odd on $[-L, L]$, then

$$a_0 = 0, \qquad a_n = 0$$

for all n, and the Fourier series becomes

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right),$$

where

$$b_n = \frac{2}{L}\int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right)\, dx.$$

Geometric and Physical Interpretation

A Fourier sine series represents an odd periodic extension of a function defined on $0 < x < L$.

$$f(-x) = -f(x)$$

This is useful when boundary values are naturally zero, such as vibrating strings fixed at both ends.

Worked Example

Example

Find the Fourier sine series of

$$f(x) = 1, \qquad 0 < x < \pi.$$

For a half-range sine expansion on $(0, \pi)$, we use

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx),$$

where

$$b_n = \frac{2}{\pi}\int_0^{\pi} 1 \cdot \sin(nx)\, dx.$$

Thus,

$$b_n = \frac{2}{\pi}\int_0^{\pi} \sin(nx)\, dx.$$

Compute the integral:

$$\int \sin(nx)\, dx = -\frac{\cos(nx)}{n}.$$

Therefore,

$$b_n = \frac{2}{\pi}\left[-\frac{\cos(nx)}{n}\right]_0^{\pi}.$$

Evaluate:

$$b_n = \frac{2}{\pi}\left[-\frac{\cos(n\pi)}{n} + \frac{\cos(0)}{n}\right].$$

Since

$$\cos(n\pi) = (-1)^n, \qquad \cos(0) = 1,$$

we get

$$b_n = \frac{2}{n\pi}\left[1 - (-1)^n\right].$$

If n is even, then $(-1)^n = 1$, so

$$b_n = 0.$$

If n is odd, then $(-1)^n = -1$, so

$$b_n = \frac{4}{n\pi}.$$

Thus,

$$1 = \frac{4}{\pi}\left[\sin x + \frac{1}{3}\sin(3x) + \frac{1}{5}\sin(5x) + \cdots\right], \qquad 0 < x < \pi.$$

Engineering Insight

Engineering Note

Fourier sine series naturally appear in vibration and wave problems where the endpoints are fixed. For example, a string fixed at both ends has displacement zero at the boundaries, and sine functions automatically satisfy this condition.

Common Mistakes and Notes

Key Concept

A Fourier sine series represents an odd extension. At $x = 0$ and $x = L$, the sine series evaluates to zero. Therefore, if the original function has nonzero endpoint values, the sine series may still represent the function on the open interval $(0, L)$ but not necessarily at the endpoints.

4 Half-Wave Symmetry

Motivation and Intuition

Many periodic engineering signals have additional symmetry beyond being even or odd. One important type is half-wave symmetry.

A waveform has half-wave symmetry if shifting it by half a period reverses its sign.

Key Concept

Let $f(x)$ be periodic with period $2L$. The function has half-wave symmetry if

$$f(x + L) = -f(x).$$

This means the second half of the waveform is the negative of the first half.

Physical Interpretation

Half-wave symmetry commonly occurs in AC waveforms, square waves, inverter outputs, and switching circuits. It implies that the positive and negative halves of the waveform balance each other.

Engineering Note

If a signal has half-wave symmetry, its average value over one period is zero. Therefore, its DC component is absent:

$$a_0 = 0.$$

Key Property

Key Concept

If $f(x)$ has half-wave symmetry, then all even harmonics vanish. That is,

$$a_{2n} = 0, \qquad b_{2n} = 0.$$

Only odd harmonics may remain.

Worked Example

Example

Consider the square wave

$$f(x) = \begin{cases} k, & 0 < x < \pi, \\ -k, & -\pi < x < 0, \end{cases}$$

with period 2π.
This function is odd, so

$$a_0 = 0, \qquad a_n = 0.$$

We compute

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\, dx.$$

Since $f(x)$ is odd and $\sin(nx)$ is odd, their product is even:

$$b_n = \frac{2}{\pi} \int_0^{\pi} k \sin(nx)\, dx.$$

Thus,

$$b_n = \frac{2k}{\pi} \int_0^{\pi} \sin(nx)\, dx.$$

Compute:

$$b_n = \frac{2k}{\pi} \left[-\frac{\cos(nx)}{n} \right]_0^{\pi}.$$

So,

$$b_n = \frac{2k}{n\pi} \left[1 - \cos(n\pi) \right].$$

Since

$$\cos(n\pi) = (-1)^n,$$

we have

$$b_n = \frac{2k}{n\pi} \left[1 - (-1)^n \right].$$

If n is even,

$$b_n = 0.$$

If n is odd,

$$b_n = \frac{4k}{n\pi}.$$

Therefore,

$$f(x) = \frac{4k}{\pi}\left[\sin x + \frac{1}{3}\sin(3x) + \frac{1}{5}\sin(5x) + \cdots\right].$$

Engineering Insight

Engineering Note

Half-wave symmetry is important in power electronics. If an inverter output has half-wave symmetry, then even harmonics are absent. This simplifies harmonic analysis and reduces certain types of distortion.

Common Mistakes and Notes

Key Concept

Half-wave symmetry is not the same as odd symmetry.
Odd symmetry means

$$f(-x) = -f(x).$$

Half-wave symmetry means

$$f(x + L) = -f(x).$$

A function may have one, both, or neither.

5 Exponential Fourier Series

Motivation and Intuition

The trigonometric Fourier series uses sine and cosine functions. However, using Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

we can rewrite Fourier series in terms of complex exponentials.

This form is especially powerful in signal processing, communications, and systems analysis.

Key Concept

For a function $f(x)$ with period L, the exponential Fourier series is

$$f(x) = \sum_{n=-\infty}^{\infty} c_n \exp\left(i\frac{2\pi nx}{L}\right),$$

where

$$c_n = \frac{1}{L}\int_0^L f(x)\exp\left(-i\frac{2\pi nx}{L}\right)dx, \qquad n \in \mathbb{Z}.$$

Interpretation

The coefficient c_n measures the contribution of the complex exponential with frequency

$$\nu_n = \frac{n}{L}.$$

Positive values of n correspond to positive frequency components, while negative values of n correspond to negative frequency components.

For real-valued signals,

$$c_{-n} = \overline{c_n}.$$

Relationship with Trigonometric Coefficients

Key Concept

For $n \geq 1$, the trigonometric coefficients and exponential coefficients are related by

$$c_n = \frac{a_n - ib_n}{2},$$

$$c_{-n} = \frac{a_n + ib_n}{2},$$

and

$$c_0 = \frac{a_0}{2}.$$

Equivalently,

$$a_n = c_n + c_{-n},$$

and

$$b_n = i(c_n - c_{-n}).$$

Worked Example

Example

Find the exponential Fourier series coefficients of

$$f(x) = \cos\left(\frac{2\pi x}{L}\right).$$

Using Euler's formula,

$$\cos\left(\frac{2\pi x}{L}\right) = \frac{1}{2}\left[\exp\left(i\frac{2\pi x}{L}\right) + \exp\left(-i\frac{2\pi x}{L}\right)\right].$$

Compare this with

$$f(x) = \sum_{n=-\infty}^{\infty} c_n \exp\left(i\frac{2\pi n x}{L}\right).$$

The term

$$\exp\left(i\frac{2\pi x}{L}\right)$$

corresponds to $n = 1$, so

$$c_1 = \frac{1}{2}.$$

The term

$$\exp\left(-i\frac{2\pi x}{L}\right)$$

corresponds to $n = -1$, so

$$c_{-1} = \frac{1}{2}.$$

All other coefficients are zero:

$$c_n = 0, \qquad n \neq \pm 1.$$

Therefore,

$$\cos\left(\frac{2\pi x}{L}\right) = \frac{1}{2}\exp\left(i\frac{2\pi x}{L}\right) + \frac{1}{2}\exp\left(-i\frac{2\pi x}{L}\right).$$

Engineering Insight

Engineering Note

The exponential Fourier series is widely used in electrical engineering because linear systems act simply on complex exponentials. If an input has Fourier coefficients c_n, then a linear time-invariant system modifies each frequency component independently.

Common Mistakes and Notes

Key Concept

Common mistakes:

1. Forgetting that the exponential series sums from $n = -\infty$ to ∞.
2. Confusing the period L with $2L$. In the exponential form above, the period is L.
3. Forgetting the negative sign in the coefficient formula:

$$c_n = \frac{1}{L} \int_0^L f(x) e^{-i2\pi nx/L} \, dx.$$

6 Filters

Motivation and Intuition

Fourier series allow us to view a signal as a collection of frequency components. A filter changes a signal by keeping, reducing, or removing selected frequencies.

Key Concept

A filter modifies the Fourier coefficients of a signal. If

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/L},$$

then a filter produces an output

$$y(x) = \sum_{n=-\infty}^{\infty} H_n c_n e^{i2\pi nx/L},$$

where H_n is the gain applied to the nth harmonic.

Physical Interpretation

A filter does not usually change all frequencies equally. Instead, it may pass low frequencies, remove high frequencies, isolate a band of frequencies, or remove unwanted oscillations.

Key Concept

Common filters:

1. Low-pass filter: keeps low frequencies and reduces high frequencies.
2. High-pass filter: keeps high frequencies and reduces low frequencies.
3. Band-pass filter: keeps frequencies within a selected range.
4. Band-stop filter: removes frequencies within a selected range.

Low-Pass Filtering

A low-pass filter keeps the slow-varying part of a signal. In Fourier series form, this means retaining small values of $|n|$ and removing large values of $|n|$.

For an ideal low-pass filter with cutoff harmonic N,

$$H_n = \begin{cases} 1, & |n| \leq N, \\ 0, & |n| > N. \end{cases}$$

Thus,

$$y(x) = \sum_{n=-N}^{N} c_n e^{i2\pi nx/L}.$$

Worked Example

Example

Suppose a periodic signal has the exponential Fourier series

$$f(x) = 3 + 2\cos\left(\frac{2\pi x}{L}\right) + \cos\left(\frac{6\pi x}{L}\right) + 0.5\cos\left(\frac{20\pi x}{L}\right).$$

Identify the harmonics:

$$3 \quad \text{is the DC component,}$$

$$2\cos\left(\frac{2\pi x}{L}\right) \quad \text{is the first harmonic,}$$

$$\cos\left(\frac{6\pi x}{L}\right) \quad \text{is the third harmonic,}$$

$$0.5\cos\left(\frac{20\pi x}{L}\right) \quad \text{is the tenth harmonic.}$$

Now pass the signal through an ideal low-pass filter that keeps harmonics up to $n = 3$. The DC, first harmonic, and third harmonic remain. The tenth harmonic is removed. Therefore, the filtered output is

$$y(x) = 3 + 2\cos\left(\frac{2\pi x}{L}\right) + \cos\left(\frac{6\pi x}{L}\right).$$

Engineering Insight

Engineering Note

Filtering is used in audio processing, communications, power electronics, instrumentation, and data analysis. For example, a rectifier converts AC to a waveform with a DC component and oscillatory components. A low-pass filter can remove the oscillatory harmonics and keep the DC value.

Connection to Average Value

The DC component is the zero-frequency part of a signal. In the exponential Fourier series, it is

$$c_0 = \frac{1}{L}\int_0^L f(x)\,dx.$$

In the trigonometric Fourier series, it corresponds to

$$\frac{a_0}{2}.$$

A perfect low-pass filter that removes all oscillatory components leaves only the average value of the signal.

$$y(x) = c_0.$$

Common Mistakes and Notes

Key Concept

Common mistakes:

1. Thinking a filter changes the time domain directly without considering frequency components.
2. Forgetting that removing high harmonics smooths the waveform.
3. Forgetting that the DC component is the average value of the signal.
4. Assuming ideal filters exist physically. Real filters approximate ideal behavior.